Trenitalia
ETR1000 VZI-6

Windscreen Heating

Revision	Author	Date	Revision Description
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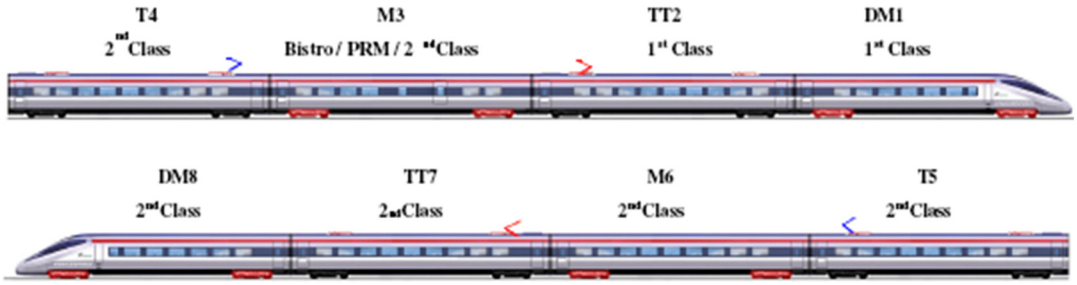
26 References 12

1 Introduction

This document is a function specification for the control of the *windscreen heating* system for the windscreen ETR1000 /VZI6 for customer Trenitalia.

2 Train Configuration

Each ETR1000 /VZI6 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

3 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System

4 System Overview

The driver's cab shall be equipped with a heatable windscreen. The windscreen heating shall be designed according TS 308474 as film heating. Additional the windscreen is equipped also with a thermostat as heating controller. If windscreen heating is out of service cab HVAC could be used as redundancy.

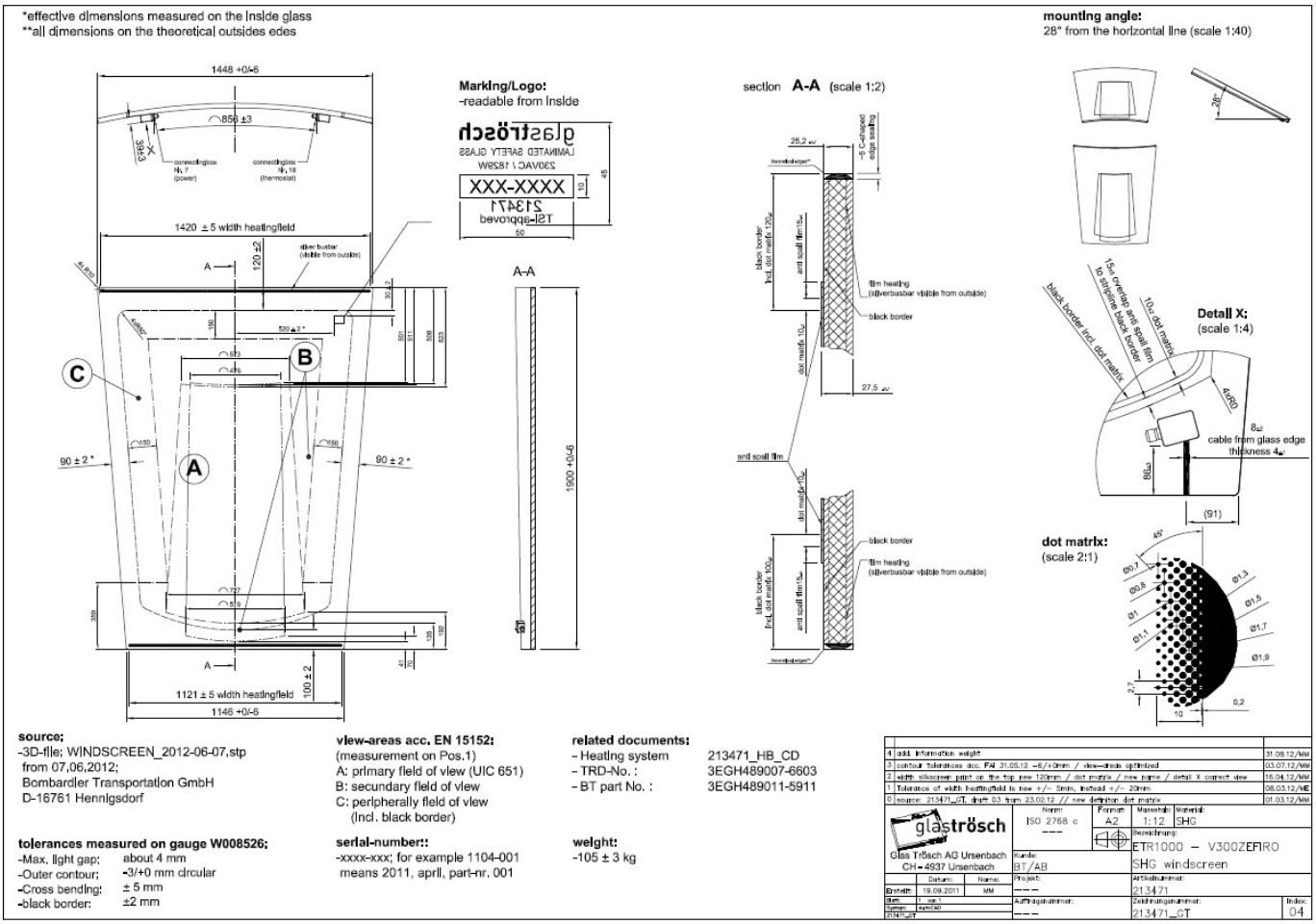
5 System Configuration

The heating operation ensured in two modes:
Mode one: Heating OFF
Mode two: Heating ON (including thermostat ON)
The stages will be activated by button on driver's desk.

In position ON the windscreen temperature shall be controlled through the thermostat on windscreen side. Is the windscreen temperature max +30°C so the heating power switched off automatically. Is the windscreen temperature min +15°C so the heating power switched on automatically when we have driver desk button in position ON. If the thermostat is damaged, the heating shall switch off automatically.

On windscreen side the power connection from heating and thermostat ends in separate connecting boxes which are bonded onto windscreen. The boxes are the interface for connecting the train side power supply. As power supply for heating are 230 VAC necessary. As power supply for thermostat are < 110 VDC necessary. Following the design principle from supplier

<Picture>



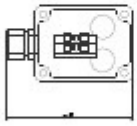
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circuit diagram heating project ETR1000 - V300ZEFIRO

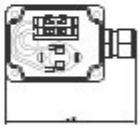
HEATING SYSTEM: - **Heating mode:** 1x230 VAC
- **Overheating control:** thermostat in control circuit

CIRCUIT DIAGRAM:

Anschlussdose/ Connnectingbox Nr 7



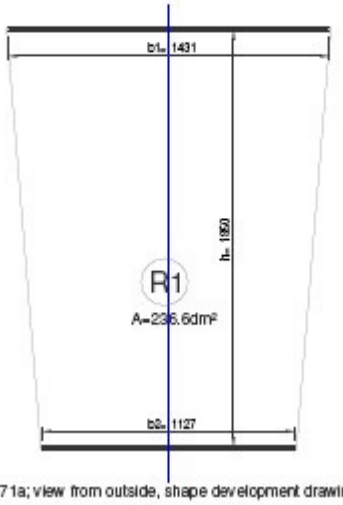
Anschlussdose/ Connectingbox Nr 18



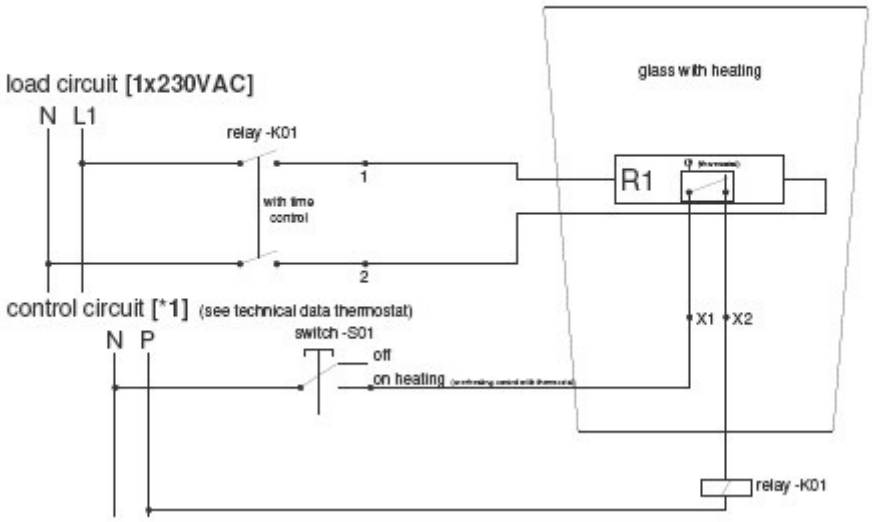
Heizen/Heating:
(pro Heizfeld/per Heatingfield)

- Spannung/Tension U:	230 VAC
- Strom/Current I:	~8.0 A
- Widerstand/Resitance R:	~28.9 Ω
- Leistung/Power:	~1829 W
- Spez. Leistung/Spec. Power:	~7.7 W/dm2





213471a; view from outside, shape development drawing



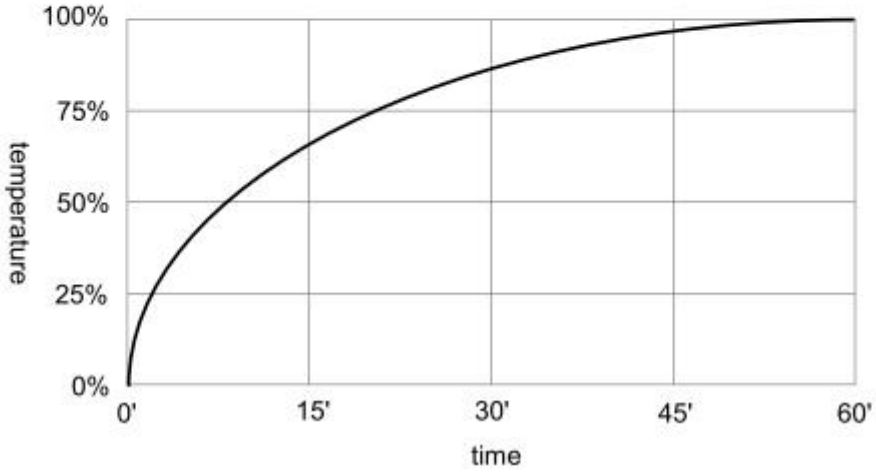
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1. Fundamentals

A heated glass will warm up about 4 – 5°C per 1 W/dm^2 specific heating rate (approximation, steady conditions, no wind). With a specific power rate of 7.7 W/dm^2 the glass may be heated up about total +35°C (4.5 x 7.7 = 34.7°C, see diagram below) within 60 minutes.

For example

Specific heating rate = 7.7 W/dm^2 about (4.5 x 7.7) = 35°C within 60 minutes (about 28°C in about 30 minutes, see diagram)



6 Functional requirements

7 Function Activation of Windscreen Heating

8 Function Design

The windscreen heating is activated by button on activated driver's desk. It's not possible to activate the heating from a not activated driver's desk. If the windscreen heating is on and the driver de-activates the desk then the windscreen heating turned off automatically. If the windscreen heating is activated, the windscreen temperature will be controlled through thermostat on windscreen side.

Regardless outdoor temperature the heating activation will be ON for 120 min, limited through TCMS. If heating is activated, on windscreen side the heating function is controlled through windscreen thermostat, if thermostat is tripped heating will be deactivated. Additional the activated heating is controlled from train side through permanent check of power supply quality including permanent check of switching status from electric contactors, relais and fuses. If heating is activated and heating has been active for 120 min it is possible to reactivate heating again by using the driver desk button. The button is equipped with light. The light is ON if heating is activated. If the heating is not activated the button light is OFF.

9 *Normal Condition*

The windscreen heating shall be activated by button on activated driver's desk.

10 *Failure cases*

1. Heating remains in OFF because button is damaged
2. Heating remains in OFF because failure in power supply
3. Heating remains in OFF because thermostat is damaged
4. Heating remains in OFF because heating is damaged

11 *Faiure detection*

1. Button light remains OFF, an event shall be stored
2. Button light remains OFF, an event shall be stored
3. Button light remains OFF, an event shall be stored
4. Button light remains OFF, an event shall be stored

12 *Failure handling*

1. Repair before train starts into service or train must break the service for repair (depends on weather conditions).
2. Repair before train starts into service or train must break the service for repair (depends on weather conditions).
3. Repair before train starts into service or train must break the service for repair (depends on weather conditions).
4. Repair before train starts into service or train must break the service for repair (depends on weather conditions). Attention: Repair means here complete window change. Change time 6 hours)

13 Requirements

ID	2F_02.03.Zefiro-Europe.TRS.21
Description	There shall be one operation modes of front window heating: - Normal Heating
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.03.Zefiro-Europe.CONCEPT.26	The heating shall be activated by the driver via a button on driver's desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.03.Zefiro-Europe.CONCEPT.244	A contactor, commanded via TCMS digital output, shall be installed to enable/disable power supply to the windscreen heating system	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_02.03.Zefiro-Europe.CONCEPT.240	When Driver activate windscreen heating pressing latched pushbutton on the driver desk, TCMS shall close the windscreen heater contactor.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_02.03.Zefiro-Europe.CONCEPT.245	As long as windscreen heating contactor is closed and 400Vac power supply is available on the train, TCMS shall lit "windscreen heating active" lamp on the driver desk (integrated inside Pushbutton)	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_02.03.Zefiro-Europe.CONCEPT.241	Windrsceen heater contactor status shall be read by TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		

2F_02.03.Zefiro-Europe.CONCEPT.242	If Contactor is not closed at order an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_02.03.Zefiro-Europe.CONCEPT.209	When heating is activated and outdoor temperature is lower than 20°C, the heating shall be kept active for 120 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.03.Zefiro-Europe.CONCEPT.161	When heating is activated and outdoor temperature is higher or equal than 20°C, the heating shall be kept active for 60 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.03.Zefiro-Europe.CONCEPT.162	If when driver activate heating the thermostat is detected tripped, an advice shall be displayed to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.03.Zefiro-Europe.CONCEPT.219	When heating is activated an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.03.Zefiro-Europe.CONCEPT.115	Heating activation shall only be possible if drivers desk is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.03.Zefiro-Europe.CONCEPT.163	Heating activation shall not be possible if drivers desk is not active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.03.Zefiro-Europe.CONCEPT.246	If heating is activated with external Temperature > 0°C, an event for maintainer shall be set in case timer is elapsed and no thermostat trip has been detected	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

14 Function De-Activation of Windscreen Heating

15 Function Design

The windscreen heating is de-activated by button on activated driver's desk. If the windscreen heating is on and the driver de-activates the desk then the windscreen heating is turned off. The windscreen heating is de-activated automatically by TCMS after 120 min heating time. The user will be informed by an event on the IDU.
Is the temperature on windscreen side +30°C so the heating power switched off automatically controlled by windscreen thermostat.

16 *Normal Conditions*

The de-activation of the heating takes place with the button on drivers desk or if time limit is reached or if thermostat is tripped (30°C threshold)

17 *Failure Cases*

1. Heating remains in ON because button is damaged
2. Heating remains in ON because thermostat is damaged

18 *Failure detection*

1. Drive detects heating activation also if pb has not been pressed
2. Diagnostic will detect missing thermostat trip (TCMS time limit added)

19 *Failure handling*

1. Repair before train starts into service or at next maintenance. During service windscreen heating will be controlled by thermostat.
2. Repair in workshop. During service windscreen heating will be controlled by thermostat.

20 Requirements

ID	2F_02.03.Zefiro-Europe.TRS.12
Description	If not de-activated manually by the driver the front window heating shall be de-activated automatically: – when ambient temperature is above threshold or – heating has been active for time period.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_02.03.Zefiro-Europe.CONCEPT.216	It shall be possible to de-activated the windscreen heating by using the button on the driver's desk or if thermostat is tripped	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.03.Zefiro-Europe.CONCEPT.204	DELETE	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.03.Zefiro-Europe.CONCEPT.205	Heating should be de-activated through TCMS (opening power supply contactor and switch off indcation) in following cases: – Push button is released by the driver – Thermostat is tripped (also if push button is pressed) – Activation timer is elapsed (see req 161) – Power supply contactor is not detected closed	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.03.Zefiro-Europe.CONCEPT.237	An event shall be set if heating is deactivated by TCMS (no driver action) including cause of dectivation	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

21 Signal Interface

22 I/O Interface

23 Supplier system

Inputs:

Outputs:

24 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.03.Zefiro-Europe.CONCEPT.224	Windscreen heating pushbutton		2F_02.03.Zefiro-Europe.CONCEPT.26 2F_02.03.Zefiro-Europe.CONCEPT.216
2F_02.03.Zefiro-Europe.CONCEPT.211	Status - Windscreen heating on/off		2F_02.03.Zefiro-Europe.CONCEPT.205

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.03.Zefiro-Europe.CONCEPT.213	Activate/deactivate windscreen heating including light in pushbutton		2F_02.03.Zefiro-Europe.CONCEPT.161 2F_02.03.Zefiro-Europe.CONCEPT.205 2F_02.03.Zefiro-Europe.CONCEPT.209

25 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.03.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.03.Zefiro-Europe.CONCEPT.33			

26 References

Supplier drawing: 213471_GT_Rev3

Supplier circuit diagram heating: 213471_HB_CD

Supplier windscreen documentation: 213471_TD